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Heterogeneous graph neural network-based prediction of immune-related adverse events

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Immune-related adverse events (irAEs) are common and potentially fatal adverse events. However, predicting irAEs based on clinical medication regimens and basic patient information remains a significant clinical challenge. This study aims to develop a prediction model using graph neural networks with electronic health records (EHRs), thereby reducing irAEs risk. Our method based on heterogeneous graph networks. It incorporates medications, diagnoses and patients characteristics from EHRs as nodes to predict irAEs occurrence. Medication-policy simulation, case studies and interpretability analyses were conducted to align the model with real-world clinical needs. Compared to other baseline methods, our method shows superior performance across all evaluation metrics: with AUC of 0.902, AUPRC of 0.85, precision of 0.709, RECALL of 0.799, F1 score of 0.751, accuracy of 0.851. About simulation study, the model demonstrated progressive improvement, reflected in a 5%–6% increase across six evaluation metrics. Interpretability analysis revealed that distinct risk patterns emerge at different treatment stages. Our approach exhibits robust reliability and outperforms other methods for irAEs prediction. Our study further establishes a novel paradigm for personalized therapy monitoring and early intervention. This methodology holds potential for reducing irAEs risk.

Keywords Immune-related adverse events, Graph neural network, Precision medicine

The emergence of immune checkpoint inhibitors (ICIs) has brought positive news to the treatment of cancer patients, significantly improving the clinical therapeutic effects of various cancers[1,2]. At present, ICIs include three types: PD-1, PD-L1, and CTLA-4. They have been approved in many countries for the treatment of various cancers, yielding substantial benefits to cancer patients[3]. Similar to other cancer treatment regimens, ICI treatment also has its limitations, namely the occurrence of immune-related adverse events (irAEs). These adverse events are diverse and vary in severity. irAEs usually triggers responses in multiple organs, such as lungs, skin, liver, heart, endocrine system, etc[4–7]. Mild irAEs may not affect the continuation of ICI treatment. However, the occurrence of severe irAEs may result in treatment interruption and pose health risks, and even lead to the death of patients[8].

With the increasing use of ICIs, current research on predicting irAEs primarily relies on collecting and integrating clinical data with conventional machine learning methods. These studies have identified factors such as age, sex, chronic disease history, and concomitant medications as potential predictors of irAEs[9–12]. However, in real-world clinical practice, ICIs are frequently administered in combination with other anticancer agents like chemotherapeutic and targeted drugs[2,13], presenting significant challenges for irAEs prediction that aligns with actual clinical practice. Furthermore, research on adverse events predominantly relies on spontaneously reported data, chemical structures, and drug molecular graphs. Most studies focus on the drug level—such as drug interactions and drug-induced suspected adverse event[14,15] with limited exploration based on clinical databases. We aim to integrate real-world clinical contexts as comprehensively as possible to develop potential methods for predicting irAEs.

At present, the electronic health records (EHRs) system is currently experiencing rapid growth. Real-world clinical data within EHRs hold significant value for pharmacovigilance, disease prevention/control, and related domains. EHRs encompass extensive, heterogeneous, and longitudinal medical information documenting patient health trajectories – including symptoms, medications, nursing interventions, and treatments.

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With the increase in EHRs data collection, people increasingly hope to conduct research on patient disease prediction, medication risk prediction, post-treatment response, etc. by mining this information, especially in the field of personalized medicine for patients[16,17]. However, how to perform reasonable data mining to obtain the results we need is a difficult problem. Undoubtedly, there are problems such as missing, irregular, and incorrect data in EHRs data[18].

Graph Neural Networks (GNNs) are deep-learning models that propagate information along edges and learn node representations[19]. By iteratively aggregating features from neighbors, GNNs simultaneously capture attribute similarity and topological structure without manual feature engineering, and naturally construct protein networks[20], gene networks[21], and patient-centric graphs that support tasks such as node classification, link prediction, or graph clustering[22]. In the realm of electronic medical records, the patients, drugs, diagnoses, and procedures contained in EHRs inherently form a heterogeneous graph whose edges encode temporal and causal dependencies that traditional tabular models struggle to capture, making GNNs especially well-suited to such data.For example, Ochoa et al.[23] constructed a patient network based on the ICD-10 codes and their meta data generated during the patient's medical process and trained a GNN model to predict the TKs (Therapy Keys) clusters during the treatment process. Zhou et al.[24] used GNN relevant methods, combined with the patient's ICD-10 diagnoses to determine the occurrence of adverse events, and constructed a network containing all ICD-10 codes of each patient for adverse event prediction. Gao et al.[25]used the FEARS reporting data, regarded the indications, medications, basic information, and adverse events of each patient as different nodes to construct an AE reporting network including all patients, and then built a model through the Heterogeneous Graph Neural Network method to achieve the prediction research of adverse reactions at the patient level. Based on the work of the above researchers, we can see that conducting relevant research by combining the characteristics of EHRs data with graph neural networks is the current mainstream method. However, for most of the data used in the above research, it can only be approximately regarded as the medical record data generated by real patients, and there are certain differences from the real data.

In this study, to predict whether irAEs occurs in the ICI-treated population, we constructed a patient care network using data from three aspects: the patient's admission diagnosis, the medication situation before the occurrence of irAEs, and the basic information. We used the Heterogeneous Graph Transformer (HGT) framework to developed a Heterogeneous Patient Graph Model (HPGM) and simulated the daily medication strategy of clinicians according to the medication time. We sorted the medication data and gradually input it into the model to achieve the prediction of irAEs at the clinical patient level in the real world. In short, the key problem we aim to address is: how to leverage real-world clinical data to predict the risk of adverse drug reactions during patient treatment.

Materials and methods

Data Preparation

All patient data used in this study were obtained from the Yiducloud Big Data Platform of the Medical Data Research Institute, Chongqing Medical University and all methods were performed in accordance with the relevant guidelines and regulations. The study was conducted in accordance with the principles outlined in the Declaration of Helsinki and was approved by the Medical Research Ethics Committee of Chongqing Medical University (2023047). All data in this database have been anonymized and structured, so this retrospective study was granted a waiver of informed consent by the ethics review board. This platform compiles patient information from seven hospitals in Southwest China. The study focused on patients treated with immune checkpoint inhibitors (ICIs) to develop a model predicting immune-related adverse events. We collected all patients who received ICIs from 2019 to 2025.

Three independent reviewers holding bachelor's degrees in medical sciences performed structured EMR audits to ascertain irAEs status for each patient. irAEs diagnoses were validated based on clinical documentation and physician assessments in medical records. The included patients were: (1) Had confirmed cancer diagnoses, (2) Received ICI therapy at participating hospitals (all ICI-treated patients were included to reflect real-world clinical scenarios). We excluded patients: Significant data incompleteness precluding reliable analysis.

The data analyzed primarily consisted of: (1) Patient demographics (age, gender, surgical history, number of hospitalizations). (2) Admission diagnoses. (3) Medication use (only drugs used before irAEs onset were included to avoid treatment - related medications).

Problem definition

To predict immune-related adverse events (irAEs) among individuals receiving ICI therapy, we developed a Heterogeneous Patient Graph Model (HPGM) by harmonizing large-scale electronic health records (EHRs) within a Heterogeneous Graph Transformer (HGT) architecture. In brief, this investigation seeks to resolve the pivotal question of how real-world, longitudinal clinical data can be systematically exploited to prospectively estimate patient-specific risk of adverse drug reactions during ICI treatment, elucidate irAEs-associated risk determinants from an EHR-centric perspective, and ultimately mitigate irAEs occurrence to ensure therapeutic safety.

Construction of HPGM model

To mimic the complex real - clinical relationships among patients, diseases, and drugs, patients, along with their related medications and diagnoses, were incorporated into a designed heterogeneous graph[25]. It included three types of nodes (patients, diseases, drugs) and edges representing the relationships between patients and medications and between patients and diagnoses. the heterogeneous graph comprises three types of nodes(patients: $\mathcal{V}_p$, drugs : $\mathcal{V}_d$, diagnoses : $\mathcal{V}_z$)and four types of edge (patient→ drug, drug→ patient, patient→

diagnosis, diagnosis→ patient), represented as: $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{\mathcal{V}_p, \mathcal{V}_d, \mathcal{V}_z\}$, the overall model framework is shown in Fig. 1.

Patient Node Features: To capture patient node characteristics, we extracted demographic information including age, gender, smoking, and alcohol history. Continuous data were standardized, and binary data were one-hot encoded. The resulting feature vectors were incorporated into the model as patient node attributes $(\mathcal{V}_p)$.

Drug and Diagnosis Node Features: For each patient, drug prescriptions and diagnostic records were chronologically ordered to reflect the temporal evolution of medical events. Each patient's clinical history thus formed a sequence of medical entities (e.g., drugs and diagnoses), which served as the basis for contextual embedding learning. Specifically, we employed a skip-gram model to learn distributed representations of these entities[26]. Within each sequence, the model predicted the surrounding context entities given a central entity, thereby capturing co-occurrence relationships that reflect medical relevance and patient-specific treatment patterns. After training, the skip-gram model produced dense vector embeddings (100 dimensions) for each unique ATC and ICD-10 code. These embeddings were used as feature representations of drug and diagnosis nodes in the heterogeneous graph.Drugs were encoded according to the Anatomical Therapeutic Chemical (ATC) classification system, while diagnoses were encoded using the International Classification of Diseases, 10th Revision (ICD-10). Both codes have classic hierarchical structures that contain medical information, and we leveraged this hierarchy to extract meaningful insights[27,28]. ATC codes were truncated to five hierarchical levels (1, 3, 4, 5 and 7 characters), corresponding to anatomical, therapeutic, pharmacological and chemical properties, respectively; ICD codes were truncated to three levels (3, 4 and 6 characters), representing different granularities. We thus obtained both semantic and hierarchical features for drug and diagnosis nodes.

For each node type, the features are mapped to a unified shared domain via the following formula to facilitate subsequent steps:

$$H_{\mathcal{V}}^{l} = \mathrm{LayerNorm}\left(H_{\mathcal{V}}^{0} W_{\mathcal{V}}^{l}\right) \tag{1}$$

where, l signifies the l-th layer.

The HGT model uses a Transformer architecture with multi-layer attention mechanisms to learn node representations, aggregating information from different types of neighboring nodes and effectively capturing

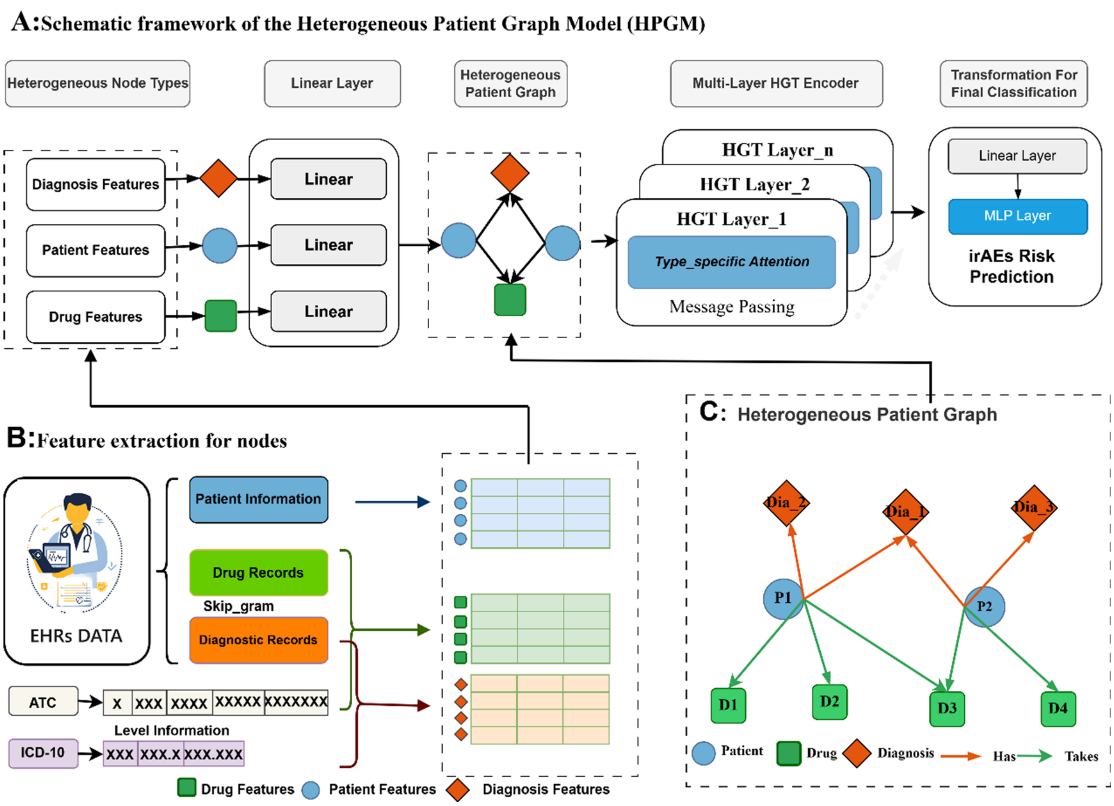

Fig. 1. (**A**) Schematic framework of the Heterogeneous Patient Graph Model (HPGM): Structured EHR data are organized into a heterogeneous graph with three node types—patients, medications, diagnoses. The HPGM iteratively updates patient representations through heterogeneous graph convolutional layers, with final predictions generated by multilayer perceptron (MLP) output modules. (**B**) Feature extraction for nodes: Skip-gram models process medication/diagnosis sequences to generate embeddings, with hierarchical structure information enhancing representation learning, both drug (ATC) and diagnosis (ICD) codes were hierarchically normalized. (**C**) Heterogeneous patient graph: in this graph it shows the basic graph structure.

complex relationships between nodes in heterogeneous graphs. Specifically, in each layer *l*, the model calculates relation-aware attention weights using the following formula:

$$\alpha_{ij}^{r} = \mathrm{softmax}\left(\frac{\left(h_i^{(l)}Q_r^K\right)\left(h_j^{(l)}Q_r^V\right)^T}{\sqrt{d_h}}\right) \tag{2}$$

Here, *ij* denotes distinct nodes, *r* represents the relationship between *i* and *j*, and *l* indicates the model's layer. By stacking multi-layer HGT convolutional layers, the model learns deep node representations. Integrating residual connections and layer normalization alleviates gradient disappearance, enhancing model generalization.

$$h_i^{(l+1)} = \mathrm{LayerNorm}\left(\sigma\left(h_i^{(l+1)}\right) + h_i^{(l)}\right) \tag{3}$$

Finally, by inputting the learned patient node representations into a Multi-Layer Perceptron (MLP) for irAEs prediction:

$$p\left(y|v\right) = \mathrm{MLP}\left(h_p^{(L)}\right) \tag{4}$$

The loss function is as follows:

$$\mathcal{L}_{\prime} = -\frac{1}{N}\sum\nolimits_{i=1}^{N} w_{y_i} \cdot \left(y_i \log\widehat{y_{i,1}} + \left(1-y_i\right)\log\widehat{y_{i,0}}\right) \tag{5}$$

N denotes the number of training samples, $y_i = \{0{,}1\}$ represents the true label, $\widehat{\mathrm{y}_{\mathrm{i},1}}$ and $\widehat{\mathrm{y}_{\mathrm{i},0}}$ re the predicted probabilities, w_{y_i} signifies the class weight.

Simulation of realistic drug administration strategies

To faithfully reflect real-world medication use, we leveraged our proposed methodology to simulate authentic drug-administration strategies. We sorted drug administration times in the database and used a one - day interval to mimic stepwise drug administration. Drugs were sequentially input into the model following this order to better reflect clinical scenarios.

Case study and interpretable analysis

To enhance model interpretability, we selected two patients with high and low risk respectively for case study. Interpretability was examined from three perspectives: (1) importance scores of the patients' baseline characteristics, (2) the dominant subgraphs that drove the individual predictions, and (3) cluster-level patterns within the patient population.

Evaluation metrics

In this study, to evaluate the predictive performance of the model, we employed six metrics: Accuracy, Precision, Recall, F1-score, AUC, and AUPRC. The computation of these metrics is as follows:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{6}$$

$$recision = \frac{TP}{TP+FP} \tag{7}$$

$$Recall = \frac{TP}{TP+FN} \tag{8}$$

$$F1 = \frac{2\times\ Precision \times\ Recall}{Precision + Recall} \tag{9}$$

$$TPR = \frac{TP}{TP+FP} \tag{10}$$

$$FPR = \frac{FP}{FP+TN} \tag{11}$$

Where, TP, TN, FP, FN, TPR, FPR, respectively, denote true positives, true negatives, false positives, true negatives, true positive rate and true positive rate. AUC denotes the area under the Receiver Operating Characteristic (ROC) curve, where the curve itself is constructed by plotting the true-positive rate (TPR) against the false-positive rate (FPR). AUPRC, in turn, is the area under the Precision–Recall curve, which is drawn with recall on the abscissa and precision on the ordinate.

Results

Patients characteristics

Following data screening through the Yidu Cloud Medical Data Platform at Chongqing Medical University, 2,576 patients were enrolled in this study. Of these, 723 (28.07%) developed immune-related adverse events (irAEs). A noteworthy majority of the patients, accounting for 1866 individuals (72.44%), were male, a total of 1,351 participants were aged over 65 years, 428 had a smoking history, and 632 had a drinking history. There were 702 individuals with emergency event records during prior treatment, and 62 had a clear record of infectious diseases. Regarding chronic disease history, 187 had a history of coronary heart disease, 326 had a history of diabetes, 668 had a history of hypertension, 243 had a history of COPD, and 271 had a history of digestive system-related diseases. Patients were grouped based on whether they experienced immune-related adverse events for chi-square analysis. The results showed that patients aged over 65, those with prior emergency events, and those with COPD or digestive system-related diseases were more prone to adverse events ($p < 0.05$), which was statistically significant (Table 1).

Model results

About baseline models in our study population, first we drew on prior studies[23,24,29,30] to build a patient-similarity network and calculated centrality metrics (like weighted degree, eigenvector, closeness, betweenness centralities, clustering coefficients, and more).We combined these with patient baseline features and used six machine learning methods: SVM, Random Forest, Logistic Regression, K - Nearest Neighbors, Decision Tree, and XGBoost. And we use the patient-similarity network for other GNN models, including GAT, GCN, and GAT2. The dataset was randomly divided into training, validation, and test sets based on a ratio of 6:2:2. The Tables 2 and 3 show the average performance of these baseline models and our model over five independent runs. The Table 4 shows the total numbers of nodes and edges in our study cohort. The Table 5 shows the

	Overall ($N = 2576$)	**Patients with irAE ($N = 723$)**	**Patients without irAE ($N = 1853$)**	***p*-value**
Sex				0.269
Female	1966(76.3)	563(77.9)	1403(75.7)	
Male	610(23.7)	160(22.1)	450(24.3)	
Age > 65				0.002
Yes	1351(52.4)	415(57.4)	936(50.5)	
No	1225(47.6)	308(42.6)	917(49.5)	
Smoking history				0.054
Yes	428(16.6)	137(18.9)	291(15.7)	
No	2148(83.4)	586(81.1)	1562(84.3)	
Alcohol history				0.675
Yes	632(24.5)	182(25.2)	450(24.3)	
No	1944(75.5)	541(74.8)	1403(75.7)	
Emergency events				< 0.001
Yes	902(35.1)	322(44.5)	580(31.3)	
No	1674(64.9)	401(55.5)	1273(68.7)	
Infectious diseases				0.145
Yes	62(2.4)	23(3.2)	39(2.1)	
No	2514(97.6)	700(96.8)	1814(97.9)	
Coronary heart diseases				0.309
Yes	187(7.3)	59(8.2)	128(6.9)	
No	2389(92.7)	664(91.8)	1725(93.1)	
Diabetes				0.429
Yes	326(12.7)	98(13.6)	228(12.3)	
No	2250(87.3)	625(86.4)	1625(87.7)	
Hypertension				0.109
Yes	668(25.9)	204(28.2)	464(25.1)	
No	1908(74.1)	519(71.8)	1389(74.9)	
COPD				< 0.001
Yes	243(9.4)	92(12.7)	151(8.1)	
No	2333(90.6)	631(87.3)	1702(91.9)	
Gastrointestinal-related diseases				0.027
Yes	271(10.5)	92(12.7)	179(9.7)	
No	2305(89.5)	631(87.3)	1674(90.3)	

Table 1. Characteristics of patients in this study.

Model	Accuracy	F1	Recall	Precision	AUC	AUPRC
Random Forest	0.741 ± 0.001	0.387 ± 0.031	0.294 ± 0.034	0.572 ± 0.027	0.706 ± 0.020	0.477 ± 0.027
Logistic regression	0.722 ± 0.020	0.232 ± 0.069	0.153 ± 0.050	0.511 ± 0.100	0.712 ± 0.017	0.480 ± 0.035
K Nearest Neighbors	0.722 ± 0.011	0.244 ± 0.043	0.161 ± 0.032	0.509 ± 0.065	0.666 ± 0.013	0.427 ± 0.011
Decision Tree	0.693 ± 0.021	0.368 ± 0.023	0.319 ± 0.015	0.436 ± 0.043	0.628 ± 0.041	0.387 ± 0.020
Extreme Gradient Boosting	0.733 ± 0.009	0.312 ± 0.030	0.217 ± 0.029	0.557 ± 0.052	0.733 ± 0.016	0.498 ± 0.019
HPGM	**0.851 ± 0.009**	**0.751 ± 0.014**	**0.799 ± 0.017**	**0.709 ± 0.020**	**0.902 ± 0.010**	**0.850 ± 0.011**

Table 2. Performances of HPGM and machine learning models.

Model	Accuracy	F1	Recall	Precision	AUC	AUPRC
GCN	0.611 ± 0.057	0.504 ± 0.023	0.706 ± 0.123	0.399 ± 0.037	0.694 ± 0.030	0.468 ± 0.023
GAT	0.390 ± 0.127	0.453 ± 0.021	**0.889 ± 0.115**	0.307 ± 0.033	0.605 ± 0.035	0.386 ± 0.049
GATv2Conv	0.764 ± 0.130	0.452 ± 0.014	0.131 ± 0.042	0.476 ± 0.111	0.619 ± 0.035	0.418 ± 0.041
HPGM	**0.851 ± 0.009**	**0.751 ± 0.014**	0.799 ± 0.017	**0.709 ± 0.020**	**0.902 ± 0.010**	**0.850 ± 0.011**

Table 3. Performances of HPGM and GNN models.

Node type	Count	Edge type	Count
Patient	2576	Patient- Drug	142,689
Drug	634	Drug- Patient	142,689
Diagnosis	1319	Patient- Diagnosis	24,205
		Diagnosis- Patient	24,205

Table 4. The total numbers of nodes and edges.

Model	Params (M)	Time per Epoch(s)	Accuracy	F1	AUC
GCN	0.02	0.83	0.61	0.50	0.69
GAT	0.28	3.42	0.39	0.45	0.61
GATv2	0.02	2.48	0.76	0.45	0.62
HPGM	0.75	1.45	0.85	0.75	0.90

Table 5. Computational efficiency of HPGM and GNN models.

computational efficiency of HPGM and GNN models. After comprehensive consideration, it can be observed that the HPGM model not only achieves the best performance across all evaluation metrics but also exhibits the highest robustness.

Ablation study

To assess each node type's contribution in our network, we conducted an ablation study. We created several ablated model versions and compared their performance to the full model. Specifically: (1) We removed all diagnosis nodes, constructing a patient-drug network (Dia_Model). (2) We removed all drug nodes, constructing a patient-diagnosis network (Drug_Model). The ablation results, summarized in the Fig. 2, indicate that both simplified models show performance drops across all metrics. This highlights that incorporating diverse information in our model enhances its performance, particularly improving recall and precision. These findings demonstrate that combining multi - faceted data strengthens the model's ability to identify the target population.

The results of realistic drug administration strategies via simulation

To align with clinical practice and enhance medication alerts for patients, we simulated real - world drug administration based on our previous research. Results showed that as drug data accumulated, model performance gradually improved. Over the 15 - day study period, AUC, F1, accuracy, recall, and precision increased by 5–6%. This indicates the model could predict outcomes even with initial drug data and improved significantly as more data was added, the results summarized in the Fig. 3, and the comparison of machine-learning models can be found in the supplementary material.

By contrast, applying the same strategy to the baseline models yields no sustained improvement, underscoring that the gains in our model stem from its capacity to exploit the progressively richer drug-administration data.

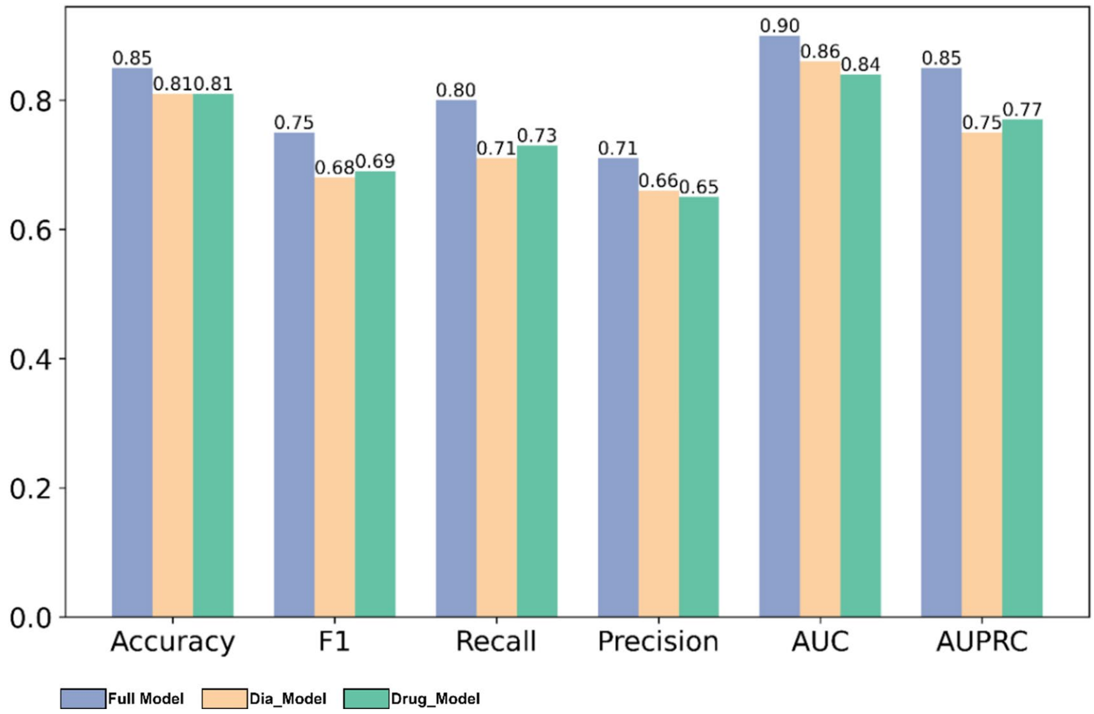

Fig. 2. The comparison of ablation results of HPGM. It highlights that the model's performance improvement is proportional to the amount of integrated information, whereas either the patient–drug network or patient–diagnosis network alone achieves significantly inferior performance.

Although the baseline Recall values diverge from ours, this discrepancy largely reflects the fluctuations induced by data shifts rather than any genuine performance enhancement of the baselines.

The 5–6% improvement in our model likely stems from its effective use of accumulating drug data to better capture patient - specific signatures of irAEs. This accumulation provides richer information for model discrimination. Our model's design enables it to dynamically update and refine predictions as new data is incorporated, which is crucial for real - world patient monitoring.

Case study and interpretable analysis

Through the above study, we have preliminarily achieved the goal of conducting prediction research based on clinical EHR data. However, for clinical research, we place greater emphasis on the interpretability of the model and its specific clinical significance. Therefore, We selected two representative patients—one who developed irAEs and one who did not—to visualize their heterogeneous subgraphs and the attention distributions learned by our model. In Fig. 4., the left side shows the original heterogeneous graphs of the two patients, while the right side shows the key subgraphs after multi-layer HGT message passing. The specific method used is Gradient-based Local Interpretation. This method measures the sensitivity of the prediction result to changes in features by calculating the gradient of the model output with respect to the input node features. For the target patient node, we backpropagate its target probability and extract gradient information related to the patient's drug and diagnosis nodes. A larger absolute value of the gradient indicates a more significant influence of that node's features on the prediction result. Finally, by visualizing the patient-drug-diagnosis subgraph, we intuitively present the model's decision-making basis at the individual patient level.

This study investigates the occurrence of irAEs by integrating electronic health records with heterogeneous graph neural networks. To enhance the interpretability of the Heterogeneous Graph Transformer (HGT) model, we address three questions at the global, individual, and subgroup levels: (1) Which patient baseline characteristics are most influential in the prediction process? (2) In an individual prediction, which nodes carry greater responsibility? (3) Do risk patterns differ across patients?

For the first question, we compute the importance of every patient-node feature with a gradient×input approach, revealing which attributes exert the strongest influence.

For the second question, we perform a case-study analysis: we visualize the target patient's most influential subgraph to identify the nodes that drive the prediction.

For the third question, after extracting the key subgraphs of all patients, we cluster them to discover subgroups that share similar feature-contribution patterns. Within each cluster we calculate the average contribution of medication and diagnosis nodes, thereby exposing characteristic differences among patient subgroups. This level of analysis helps identify sub-populations with comparable drug-risk profiles and provides a basis for stratified clinical intervention.

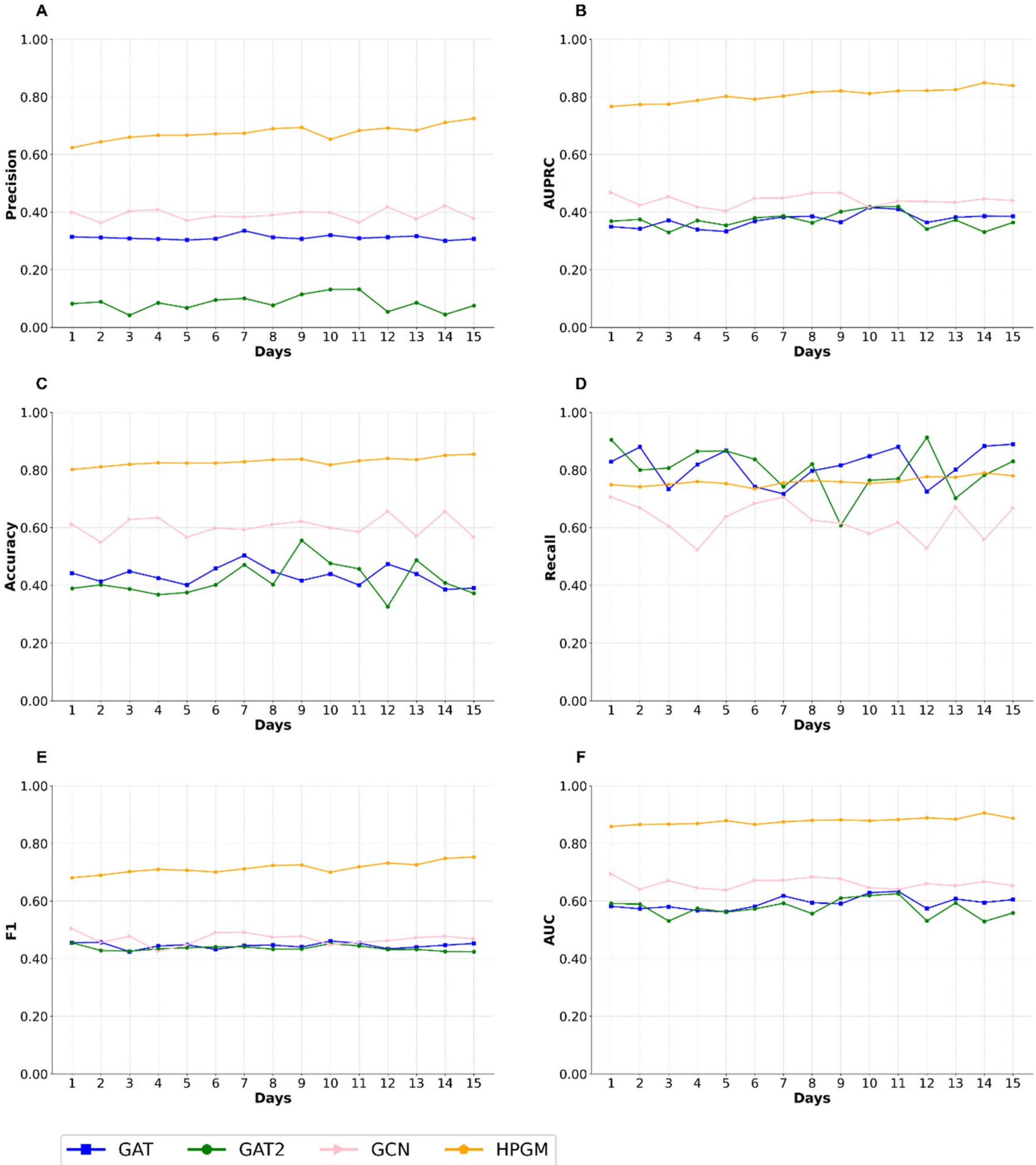

Fig. 3. Performance of Models Under Real Dosing Strategy. It demonstrated that, compared to other models, the performance of our model improves as more information is incorporated.

Concrete inspection shows that, for Patient A in Fig. 4, the influential subgraph centers on non-cancer medications—agents aimed at relieving discomfort and improving quality of life—indicative of a palliative-care phase. In contrast, Patient B's subgraph highlights cancer-directed therapeutics, reflecting an active-treatment phase. This suggests that patients at different stages may warrant distinct considerations.

In Fig. 5.regarding diagnoses (Fig. A and C), the irAEs group was dominated by malignant neoplasms and their metastatic foci: esophageal malignancy (contribution ≈ 0.10), secondary malignancy of unspecified site (≈ 0.08), and secondary malignancy of liver/intra-hepatic bile ducts (≈ 0.07). In contrast, the non-irAEs group was characterized by palliative care (≈ 0.24), metastatic malignant disease (≈ 0.20), and admission for antineoplastic chemotherapy plus symptomatic treatment. For medications (Fig. B and D), the irAEs group—besides supportive agents such as sodium chloride (≈ 0.48) and omeprazole (≈ 0.28)—showed the highest contributions from immunomodulators: epoetin alfa (≈ 0.42) and thymopentin (≈ 0.35). The non-irAEs group was led by sodium chloride (≈ 0.55), followed by nivolumab (≈ 0.38), antibacterials (piperacillin-tazobactam, ≈ 0.30), and chemotherapy (cisplatin, ≈ 0.25). These findings demonstrate a consistent pattern: the irAEs group is predominantly characterized by chemo-immunotherapy combinations, whereas the non-irAEs group leans toward regimens focused on maintaining quality of life.

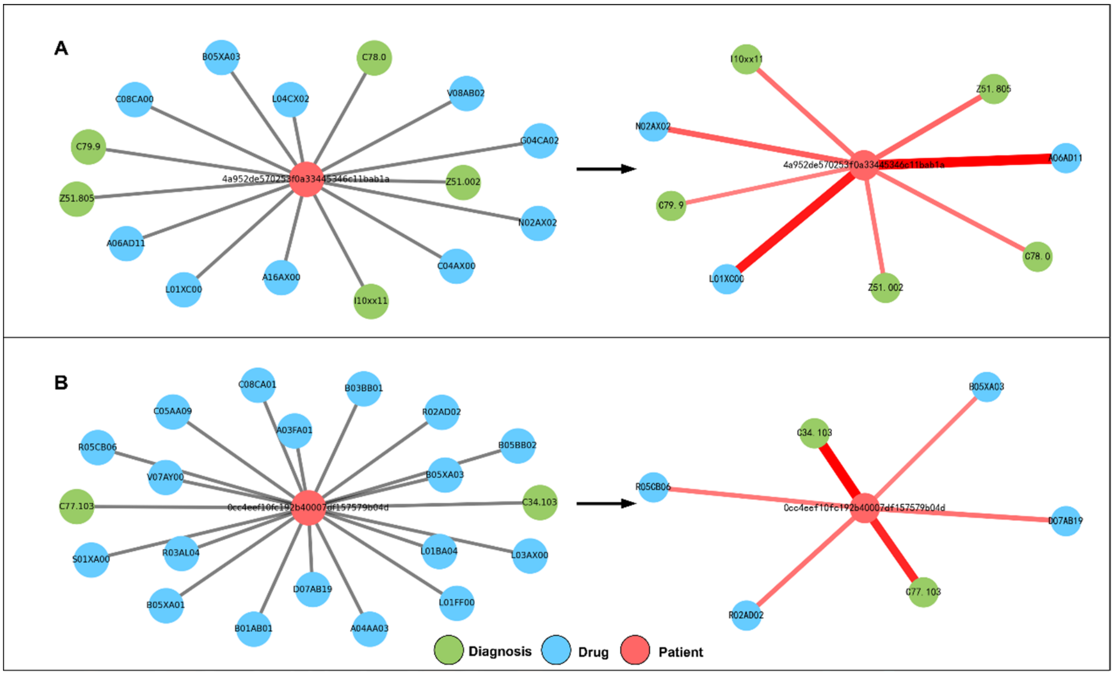

Fig. 4. Case analysis figure. (**A**) This patient has a low risk of developing irAEs; (**B**) This patient has a high risk of developing irAEs. The left panel presents the original heterogeneous graph of each patient, and the right panel displays the key subgraphs after multi-layer HGT message passing.

Finally, feature-importance analysis of patient nodes reveals that a more complex prior treatment history is associated with higher irAEs risk (Fig. 6.).

Taken together, our preliminary conclusion is that patients with extensive prior medical complexity and aggressive current treatment regimens merit heightened attention because they carry the greatest risk.

Discussion

The widespread adoption of EHRs has demonstrated their substantial utility in medical research. Our study enrolled 2,576 patients and initially conducted a preliminary analysis of common factors associated with irAEs. We identified that age and pre-existing health conditions may influence irAEs occurrence in our cohort, aligning with prior research findings.

Our study aims to predict irAEs in real - clinical settings. By integrating graph neural network techniques with EHRs data characteristics, we ensure the model's practicality. Compared to traditional machine learning methods, our model shows superior performance and better alignment with actual medication administration strategies. It provides a new approach for personalized healthcare, meeting the demand for continuous and frequent medication monitoring in clinical settings and embracing the patient - centered care philosophy[31].

Given the data features included in our study, we adopted rational approaches to learn potential information and representations from drug and diagnosis nodes, informed by prior EHR research[32]. This enabled us to build a patient - level irAEs prediction framework. Experimental results validate the effectiveness of our method in leveraging diverse medical - process - generated data for target prediction. Specifically, our method demonstrates significant performance improvements across all evaluation metrics compared to traditional machine learning prediction methods.

Ablation studies further demonstrate the importance of incorporating a diverse range of data into the model. Simplifying the model leads to reduced predictive performance, highlighting the benefits of using comprehensive data in irAEs prediction. Additionally, by considering the sequential nature of drug administration in real - world clinical practice, we transformed drug - related data into sequential data and incrementally incorporated it into our model. This approach allows our model to dynamically adjust predictions based on changing medication regimens, better meeting the expectations of clinical staff for prediction models that can adapt to real - time changes in patient conditions and treatments.To enhance interpretability and clinical utility, we conducted case-level interrogation and model-level post-hoc analyses. After mapping salient subgraph nodes to standardised clinical terminologies, we found that the HGT signal was predominantly driven by prior morbidity burden and current treatment phase, implicating these factors as the mechanistic underpinnings of irAEs risk captured by the EHR-based model.

This research represents a significant advancement in precision medicine, distinct from prior EHR-based studies through its integration of medical knowledge and clinical workflow context[33,34]. We believe our research will have a far - reaching impact on EHRs data studies. Its applications extend beyond pharmacovigilance to other areas such as disease and health - status prediction, enabling more patient - focused care and providing suitable health recommendations for each individual.

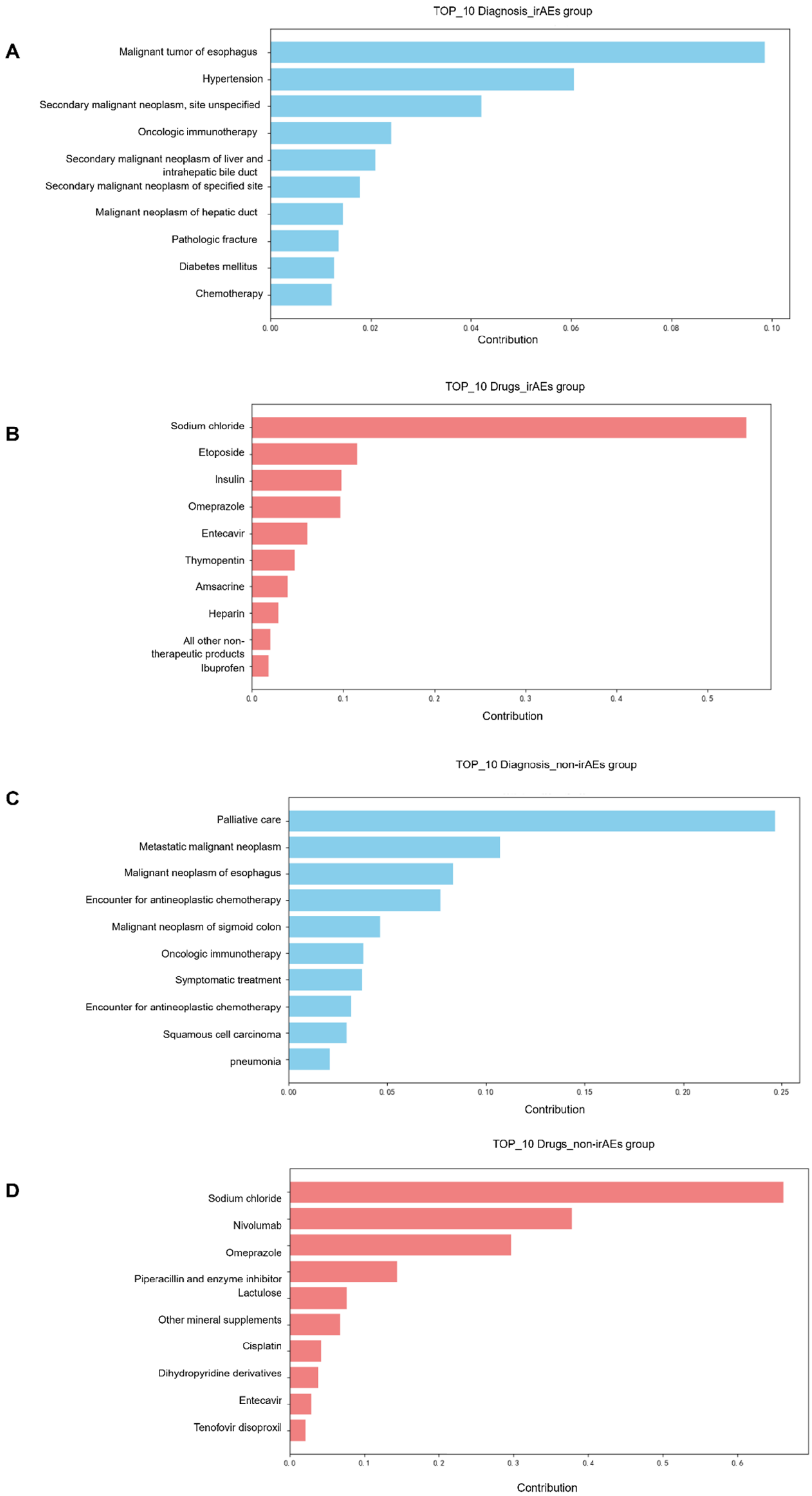

Fig. 5. cluster analysis **of** diagnoses and drugs in immune-related and non-immune-related adverse events (**A**) Top 10 diagnoses contributing to irAEs; (**B**) Top 10 drugs contributing to irAEs; (**C**) Top 10 diagnoses contributing to non- irAEs; (**D**) Top 10 drugs contributing to non- irAEs. The x-axis represents the contribution value, and the y-axis lists the diagnoses or drugs.

Despite the demonstrated effectiveness of our method in predicting immune - related adverse events, we acknowledge its limitations. First, our data originates from real - world medical treatment processes, making errors and omissions inevitable. Currently, our study primarily addresses whether a patient will develop irAEs, but further research is needed to predict the specific types of irAEs that may occur.Another consideration is

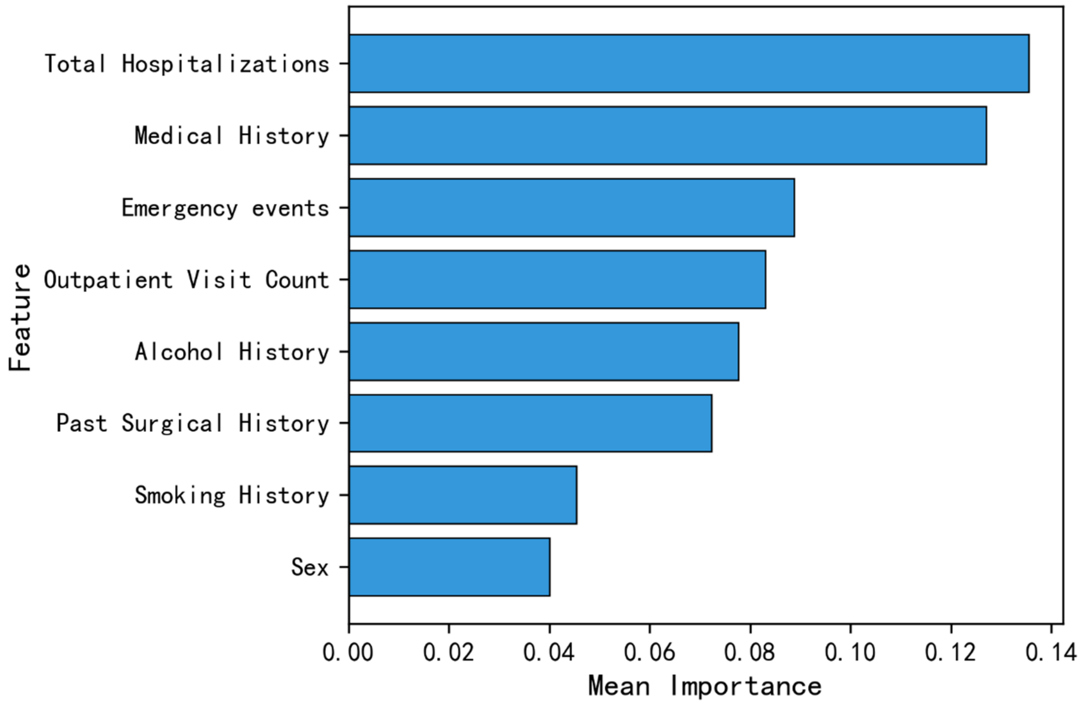

Fig. 6. Patient base features importance.

the scalability of our approach. In the future, we aim to incorporate more diverse data sources, such as disease - related genes, drug targets, and drug - associated proteins. This will enrich our network and enhance its comprehensiveness. We also plan to include more prior knowledge, such as drug - therapy information for specific diseases, drug similarities, and disease similarities, as well as other nodes relevant to drug - induced adverse reactions[27,35–37].

Conclusion

Our study introduces a novel approach for personalized immune - related adverse reaction prediction. Compared to traditional adverse - event prediction models[38,39], our method achieves superior performance across all metrics. By utilizing real - hospital data and focusing on the relationships between patients, drugs, and diseases, we have constructed a model that advances the development of precision medicine. Our approach emphasizes patient - specific characteristics and medication use, offering valuable insights for clinical practice. Furthermore, by integrating real-world medication-administration strategies with mechanistic insights derived from our interpretability pipeline, the model can provide clinicians with actionable recommendations for individualized prescribing, thereby mitigating adverse events and safeguarding patient safety.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to privacy and ethical considerations. However, interested parties may request access to the data from the corresponding author.

Received: 15 September 2025; Accepted: 24 November 2025
Published online: 29 November 2025

References

1. Gou, M., Zhang, Y., Wang, Z., Qian, N. & Dai, G. PD-1 inhibitor combined with albumin Paclitaxel and apatinib as second-line treatment for patients with metastatic gastric cancer: a single-center, single-arm, phase II study. *Invest. New. Drugs*. **42**, 171–178. https://doi.org/10.1007/s10637-024-01425-3 (2024).
2. Zhu, C. et al. Efficacy, safety, and prognostic factors of PD-1 inhibitors combined with lenvatinib and Gemox chemotherapy as first-line treatment in advanced intrahepatic cholangiocarcinoma: a multicenter real-world study. *Cancer Immunol. Immunother.* **72**, 2949–2960. https://doi.org/10.1007/s00262-023-03466-8 (2023).
3. Naimi, A. et al. Tumor immunotherapies by immune checkpoint inhibitors (ICIs); the pros and cons. *Cell. Commun. Signal.* **20**, 44. https://doi.org/10.1186/s12964-022-00854-y (2022).
4. Wright, J. J., Powers, A. C. & Johnson, D. B. Endocrine toxicities of immune checkpoint inhibitors. *Nat. Rev. Endocrinol.* **17**, 389–399. https://doi.org/10.1038/s41574-021-00484-3 (2021).
5. Chang, C-Y. et al. Immune checkpoint inhibitors and immune-related adverse events in patients with advanced melanoma: a systematic review and network meta-analysis. *JAMA Netw. Open.* **3**, e201611. https://doi.org/10.1001/jamanetworkopen.2020.1611 (2020).

6. Gougis, P. et al. Clinical spectrum and evolution of immune-checkpoint inhibitors toxicities over a decade-a worldwide perspective. *EClinicalMedicine* **70**, 102536. https://doi.org/10.1016/j.eclinm.2024.102536 (2024).
7. Ramos-Casals, M. et al. Immune-related adverse events of checkpoint inhibitors. *Nat. Rev. Dis. Primers.* **6**, 38. https://doi.org/10.1038/s41572-020-0160-6 (2020).
8. Barroso-Sousa, R. et al. Endocrine dysfunction induced by immune checkpoint inhibitors: practical recommendations for diagnosis and clinical management. *Cancer* **124**, 1111–1121. https://doi.org/10.1002/cncr.31200 (2018).
9. Gu, T. et al. Adverse reactions associated with immune checkpoint inhibitors and bevacizumab: A pharmacovigilance analysis. *Int. J. Cancer*. **152**, 480–495. https://doi.org/10.1002/ijc.34332 (2023).
10. McQuade, J. L. et al. Association of body mass index with the safety profile of nivolumab with or without ipilimumab. *JAMA Oncol.* **9**, 102–111. https://doi.org/10.1001/jamaoncol.2022.5409 (2023).
11. Cortellini, A. et al. Another side of the association between body mass index (BMI) and clinical outcomes of cancer patients receiving programmed cell death protein-1 (PD-1)/ programmed cell death-ligand 1 (PD-L1) checkpoint inhibitors: A multicentre analysis of immune-related adverse events. *Eur. J. Cancer*. **128**, 17–26. https://doi.org/10.1016/j.ejca.2019.12.031 (2020).
12. Gong, L. et al. Identification and prediction of immune checkpoint inhibitors-related pneumonitis by machine learning. *Front. Immunol.* **14**, 1138489. https://doi.org/10.3389/fimmu.2023.1138489 (2023).
13. Kao, M-W. et al. Immune checkpoint inhibitor, nivolumab, combined with chemotherapy improved the survival of unresectable advanced and metastatic esophageal squamous cell carcinoma: a real-world experience. *Int. J. Mol. Sci.* **24**, 7312. https://doi.org/10.3390/ijms24087312 (2023).
14. Han, X., Xie, R., Li, X., Li, J. & SmileGNN Drug-drug interaction prediction based on the smiles and graph neural network. *Life (Basel)*. **12**, 319. https://doi.org/10.3390/life12020319 (2022).
15. Wang, J. et al. Multi-view feature representation and fusion for drug-drug interactions prediction. *BMC Bioinform.* **24**, 93. https://doi.org/10.1186/s12859-023-05212-4 (2023).
16. Suo, Q. et al. Deep patient similarity learning for personalized healthcare. *IEEE Trans. Nanobiosci.* **17**, 219–227. https://doi.org/10.1109/TNB.2018.2837622 (2018).
17. Zhang, D., Yin, C., Zeng, J., Yuan, X. & Zhang, P. Combining structured and unstructured data for predictive models: a deep learning approach. *BMC Med. Inf. Decis. Mak.* **20**, 280. https://doi.org/10.1186/s12911-020-01297-6 (2020).
18. Si, Y. et al. Deep representation learning of patient data from electronic health records (EHR): A systematic review. *J. Biomed. Inf.* **115**, 103671. https://doi.org/10.1016/j.jbi.2020.103671 (2021).
19. Veličković, P. Everything is connected: graph neural networks. *Curr. Opin. Struct. Biol.* **79**, 102538. https://doi.org/10.1016/j.sbi.2023.102538 (2023).
20. Yang, Y. et al. FMvPCI: A multiview fusion neural network for identifying protein complex via fuzzy clustering. *IEEE Trans. Syst. Man. Cybern Syst.* **55**, 6189–6202. https://doi.org/10.1109/TSMC.2025.3578348 (2025).
21. Su, X. et al. Interpretable identification of cancer genes across biological networks via transformer-powered graph representation learning. *Nat. Biomed. Eng.* **9**, 371–389. https://doi.org/10.1038/s41551-024-01312-5 (2025).
22. Yang, Y. et al. Link-based attributed graph clustering via approximate generative bayesian learning. *IEEE Trans. Syst. Man. Cybern Syst.* **55**, 5730–5743. https://doi.org/10.1109/TSMC.2025.3572738 (2025).
23. Diaz Ochoa, J. G. & Mustafa, F. E. Graph neural network modelling as a potentially effective method for predicting and analyzing procedures based on patients' diagnoses. *Artif. Intell. Med.* **131**, 102359. https://doi.org/10.1016/j.artmed.2022.102359 (2022).
24. Zhou, F., Khushi, M., Brett, J. & Uddin, S. Graph neural network-based subgraph analysis for predicting adverse drug events. *Comput. Biol. Med.* **183**, 109282. https://doi.org/10.1016/j.compbiomed.2024.109282 (2024).
25. Gao, Y. et al. Precision adverse drug reactions prediction with heterogeneous graph neural network. *Adv. Sci. (Weinh)*. **12**, e2404671. https://doi.org/10.1002/advs.202404671 (2024).
26. Hollis, G., Westbury, C. & Lefsrud, L. Extrapolating human judgments from skip-gram vector representations of word meaning. *Q. J. Exp. Psychol. (Hove)*. **70**, 1603–1619. https://doi.org/10.1080/17470218.2016.1195417 (2017).
27. Jia, Z., Lu, X., Duan, H. & Li, H. Using the distance between sets of hierarchical taxonomic clinical concepts to measure patient similarity. *BMC Med. Inf. Decis. Mak.* **19**, 91. https://doi.org/10.1186/s12911-019-0807-y (2019).
28. Nickel, J. et al. SuperPred: update on drug classification and target prediction. *Nucleic Acids Res.* **42**, W26–31. https://doi.org/10.1093/nar/gku477 (2014).
29. Lu, H., Uddin, S., Hajati, F., Moni, M. & Khushi, M. A patient network-based machine learning model for disease prediction: the case of type 2 diabetes mellitus. *Appl. Intell.* **52**, 2411–2422. https://doi.org/10.1007/s10489-021-02533-w (2022).
30. Xiao, C. et al. Sarcopenia prediction using multi-modal data fusion and weighted graph attention networks. *Proc. Symp. Appl. Comput.* **2023**, 614–617. https://doi.org/10.1145/3555776.3578731 (2023).
31. Zhang, S., Bamakan, S. M. H., Qu, Q. & Li, S. Learning for personalized medicine: a comprehensive review from a deep learning perspective. *IEEE Rev. Biomed. Eng.* **12**, 194–208. https://doi.org/10.1109/RBME.2018.2864254 (2019).
32. Kwak, H. et al. Drug-disease graph: predicting adverse drug reaction signals via graph neural network with clinical data. *Adv. Knowl. Discovery Data Min.* **12085**, 633–644. https://doi.org/10.1007/978-3-030-47436-2_48 (2020).
33. Yang, Z. et al. Using deep learning and electronic health records to detect Noonan syndrome in pediatric patients. *Genet. Med.* **24**, 2329–2337. https://doi.org/10.1016/j.gim.2022.08.002 (2022).
34. Jiang, X. & Xu, C. Deep learning and machine learning with grid search to predict later occurrence of breast cancer metastasis using clinical data. *J. Clin. Med.* **11**, 5772. https://doi.org/10.3390/jcm11195772 (2022).
35. Zeng, X. et al. Measure clinical drug-drug similarity using electronic medical records. *Int. J. Med. Inf.* **124**, 97–103. https://doi.org/10.1016/j.ijmedinf.2019.02.003 (2019).
36. Farnoush, A., Sedighi-Maman, Z., Rasoolian, B., Heath, J. J. & Fallah, B. Prediction of adverse drug reactions using demographic and non-clinical drug characteristics in FAERS data. *Sci. Rep.* **14**, 23636. https://doi.org/10.1038/s41598-024-74505-2 (2024).
37. Pirmohamed, M. Personalized pharmacogenomics: predicting efficacy and adverse drug reactions. *Annu. Rev. Genomics Hum. Genet.* **15**, 349–370. https://doi.org/10.1146/annurev-genom-090413-025419 (2014).
38. Chen, J., Han, F., He, M., Shi, Y. & Cai, Y. A novel weighted pseudo-labeling framework based on matrix factorization for adverse drug reaction prediction. *BMC Bioinform.* **26**, 54. https://doi.org/10.1186/s12859-025-06053-z (2025).
39. Chen, Y-H., Shih, Y-T., Chien, C-S. & Tsai, C-S. Predicting adverse drug effects: A heterogeneous graph Convolution network with a multi-layer perceptron approach. *PLoS One*. **17**, e0266435. https://doi.org/10.1371/journal.pone.0266435 (2022).

Acknowledgements

We thank all the participants, as no meaningful research could have been conducted without them.And we extend our gratitude to Yiducloud Big Data Platform for its data support.

Author contributions

Xiaojun He: Writing – review & editing, Writing-original draft, Validation, Methodology, Investigation, Formal analysis and data review; Qiao Ni: Writing – review & editing, Formal analysis, investigation and data review; Cui Chen: Data curation; Hongmei Li: Formal analysis; Linghao Ni: Software, writing-review; Jiawei Zhou, Lan Tang and Bin Peng: Conceptualization and writing-review & editing. All the authors contributed to the article and approved the final manuscript.

Funding
This study was supported by grants from the National Natural Science Foundation of China (No.82273739).

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-30266-0.

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